

TORPABRON, Sweden

WMO: 025150

Lat: 58.1997N

Lon: 12.1541E

Elev: 56

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.8	8.0	-0.7	5.3	2.2	5.3	7.3	9.2	31.6	46.2	27.6	44.3	0.8	90	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	81.0	65.7	77.5	64.6	73.8	62.6	68.0	77.0	66.2	73.9	64.4	71.4	7.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	92.3	70.7	63.2	86.9	69.1	61.6	82.1	67.5	32.3	77.6	30.9	74.5	29.5	71.4	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	20.1	17.4	DB	-4.8	85.6	9.7	2.9	-11.8	87.6	-17.5	89.3	-22.9	91.0	-29.9	93.1	(4)
(5)			WB	-5.3	70.2	8.4	2.3	-11.4	71.9	-16.3	73.2	-21.1	74.5	-27.2	76.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	28.6	30.1	34.9	43.3	51.5	57.7	62.9	61.4	54.0	45.5	39.3	32.3	(6)
(7)		DBStd	13.70	10.40	8.62	7.60	5.60	5.52	4.58	4.31	4.37	4.88	6.86	7.37	10.23	(7)
(8)		HDD50	3008	662	557	467	210	45	2	0	0	20	171	325	549	(8)
(9)		HDD65	7272	1127	977	932	650	419	224	95	127	331	605	771	1014	(9)
(10)		CDD50	1261	0	0	0	10	91	233	399	354	139	31	4	0	(10)
(11)		CDD65	50	0	0	0	0	0	5	30	15	0	0	0	0	(11)
(12)		CDH74	695	0	0	0	0	17	145	376	153	4	0	0	0	(12)
(13)		CDH80	111	0	0	0	0	1	24	72	14	0	0	0	0	(13)
(14)	Wind	WSAvg	7.2	7.9	7.5	7.5	7.1	8.0	7.1	6.5	6.2	7.5	6.6	7.7	7.3	(14)
(15)	Precipitation	PrecAvg	37.2	3.4	2.6	2.2	2.2	2.2	3.1	3.3	3.5	3.6	4.1	3.5	3.5	(15)
(16)		PrecMax	48.9	6.7	6.4	4.8	4.8	4.1	6.4	6.9	7.0	8.5	8.2	7.0	8.1	(16)
(17)		PrecMin	26.6	0.4	0.7	0.5	0.4	0.4	0.3	0.2	0.3	0.8	1.1	1.1	0.6	(17)
(18)		PrecStd	5.9	1.8	1.5	1.1	1.3	0.9	1.4	1.8	1.5	1.8	1.8	1.7	2.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.6	48.4	58.4	68.0	75.9	83.4	86.2	81.9	74.1	63.2	56.5	51.5	(19)
MCWB			45.3	46.0	47.3	53.8	59.1	64.2	68.7	66.9	62.8	56.0	54.2	49.7	(20)	
2%		DB	45.3	45.2	52.1	63.6	71.5	78.1	82.4	78.5	68.9	59.4	53.8	49.2	(21)	
		MCWB	43.2	42.5	44.3	51.0	57.7	62.2	67.2	66.1	59.9	55.2	51.4	47.5	(22)	
5%		DB	43.5	43.4	47.9	59.6	67.6	73.9	78.8	74.6	65.7	57.5	51.7	47.4	(23)	
		MCWB	41.5	41.3	42.0	49.1	56.0	60.4	65.8	64.7	58.3	54.3	49.5	45.5	(24)	
10%		DB	41.7	41.6	44.8	55.4	62.9	69.4	74.6	71.0	62.9	55.7	50.0	45.7	(25)	
		MCWB	40.0	39.6	40.9	47.0	53.2	58.7	64.2	62.7	57.0	52.9	48.0	43.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.8	46.6	47.8	54.6	62.8	67.1	70.6	70.1	63.9	58.6	54.6	50.1	(27)
MDCB			47.0	48.2	56.2	64.8	71.0	76.1	80.9	76.0	70.4	60.4	55.9	50.9	(28)	
2%		WB	43.7	43.3	45.4	52.5	59.4	63.9	68.7	67.5	61.6	56.4	51.7	48.0	(29)	
		MDCB	45.0	44.7	50.5	61.4	67.4	73.9	79.0	74.8	66.6	58.3	53.2	49.0	(30)	
5%		WB	41.8	41.5	43.2	50.0	57.3	62.1	67.0	65.7	59.8	54.8	49.9	45.6	(31)	
		MDCB	43.1	43.0	46.7	56.9	64.4	70.9	75.9	72.4	63.9	56.9	51.6	47.0	(32)	
10%		WB	40.2	39.7	41.7	47.9	55.1	60.1	65.2	64.1	58.3	53.3	48.0	43.7	(33)	
		MDCB	41.5	41.5	44.3	54.2	61.7	67.7	72.4	70.1	61.8	55.6	49.7	45.4	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.4	10.3	14.5	18.5	17.7	18.0	17.2	16.8	15.8	13.5	8.9	9.0	(35)
MCDBR			9.4	9.0	19.4	28.2	26.0	28.4	27.0	23.7	21.3	14.1	9.1	8.1	(36)	
MCWBR			9.1	7.2	12.4	16.2	13.1	13.6	13.1	12.4	12.8	10.1	8.5	7.9	(37)	
5% WB		MCDBR	9.2	8.2	16.6	23.3	21.1	24.0	22.4	20.7	17.6	12.0	8.6	7.8	(38)	
		MCWBR	8.9	7.1	11.4	14.2	11.9	12.6	12.1	11.9	11.7	9.4	8.3	7.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.294	0.325	0.343	0.382	0.377	0.366	0.388	0.386	0.354	0.342	0.313	0.278	(40)	
taud		2.243	2.307	2.351	2.295	2.380	2.436	2.395	2.400	2.476	2.414	2.298	2.184	(41)		
Ebn at Noon		172	223	254	261	272	277	267	258	250	216	163	137	(42)		
Edh at Noon		15	23	28	35	34	33	34	31	25	20	14	11	(43)		
(44)	All-Sky Solar Radiation	RadAvg	110	293	765	1250	1659	1853	1678	1317	894	405	144	76	(44)	
(45)		RadStd	21	50	74	159	143	109	195	107	101	62	27	8	(45)	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (56 neighbors)	+0.70	N/A	N/A	N/A	N/A	-227	-288	N/A	N/A	(47)

Nomenclature: See separate page